

NoteBlock: Prototype Design of Music Learning Experience for Blind and Low-Vision Children in Preschool Ages

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Abstract

This research aims to create an interactive music learning device, NoteBlock, for Blind or Low Vision (BLV) children. Through research on music learning-related designs and interactive designs suitable for BLV children, NoteBlock proposes a prototype for preschool children to understand music notes and motivate their interest in music. This research aims to investigate the mental model of BLV children through feasibility testing, primarily enhancing user experience based on the design concepts of ergonomics and universal design. Overall, NoteBlock proved engaging and effective interaction in music education. Further improvements suggested a broader usability.

CCS C Concepts

Human-centered computing • Human-centered computing • Human-computer interaction (HCI) • Interaction devices

Keywords

Music Learning, Modular Interactive Design, Blind or Low Vision (BLV), Preschool-aged Children

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1 Introduction

Music can motivate self-development, creative expression, and community building for blind or low vision (BLV) individuals[1]. According to McKinnon's study on early music education for children, music learning enhances higher brain function and improves reading and math performances[2]. Emile Jaques-Dalcroze discussed that bodily and musical processes are linked[3]. BLV children rely on physical perception with a particular focus on developing their tactile and musical awareness. Music education enhances children's auditory perception, elevates sensitivity to sound and rhythm, and enriches emotional expression and experience. According to the Music Curriculum Standards for Compulsory Education in Blind Schools formulated by the Ministry of Education of China, the primary music teaching objective in the early stages of blind schools entails leveraging auditory and tactile experiences to fuel curiosity, employing an integrated teaching approach that combines listening, singing, and body movement, while also incorporating game elements to attain the set goals [4]. Therefore, we aim to create a device suitable for school and home use. Early childhood is a critical stage of development, during which experiences may affect long-term development outcomes[5]. However, the resources for blind schools in China can be limited, and it is not easy to purchase expensive musical instruments in the early stages of learning. In addition, whether parents cooperate with teachers or act alone, they play a decisive role in enriching children's learning experiences and processes[6]. Therefore, we proposed an affordable solution for both school and family use that can encourage young children to engage in music learning. The observational and interview research is arranged with three teachers from two special schools to obtain relevant research directions and content suggestions. Two partially blind children were invited to complete the initial feasibility tests of these prototypes.

2 Related Works

Through research on music interaction design studies, we have found that variation of stimulation can enrich children's

perceptual abilities and improve their music learning experience. In addition, some studies have also revealed the cognitive connection between music and programming.

2.1 Enrich Sensation by Multiple Stimulation

The case study of related work starts with the gamified interactive designs of music visualization. For example, an interactive visual music tool with synesthesia between music and vision was developed to involve scale, interval and harmony[7], and a prototype of a music visualization system can collect played music data and transform it into physical stimulations such as movements, vibrations, and lighting signals[8]. Although visualizing music can provide a fun experience, it is not suitable for BLV children. As Milne et al. proposed using the touch screen and vibration motor on smartphones to learn Braille[9], McElligott and Leeuwen also highlight the integration of the audio and tactile sensory experience into the computer game[10]. In designs related to music composition and performance, Crea8tor uses attached accelerometer sensors to provide control[11], while Lu et al. believe that Musical Haptic Wearables with vibrotactile alerts and variations in vibration are suited for BLV audiences to communicate information[12]. Besides, the hearing sense of BLV users is often better than other users, so Dang et al. attempted to improve the music performance experience through Non-Visual VR interaction[13], and Sánchez et al. also defined the usability of haptic devices and sound-based videogames for school-age children[14]. However, Ouyang et al. discussed the conversational agents in interactive speakers to provide music education to Preschoolers, multiple factors can lead to communication breakdowns[16], and design should simplify operational methods and communication feedback patterns for younger BLV users.

2.2 Studies in Music Learning Experiences for BLV Children

McDowall researched how computer-based music technology and programs can help children with the complex processes of music learning by using new forms that link sounds and visual symbols[17]. Many interactive designs that discuss the relationship between programming (or coding) and music learning use modular presentation. For instance, Peterson and Haynes argued that coding is a new style of music education that focuses on composition and structural description instead of the performance of existing music[18]. Therefore, Villar et al. proposed a physical programming language for children ages 7-11 to create music, audio stories or poetry by connecting and manipulating tactile objects[1]. In addition, Sabuncuoglu also developed an affordable and accessible tangible music platform that uses basic programming ideas for music creation[19], and it is similar to Fischer and Lau's interactive model that works on sequencers for music education through haptic programming

ideas[20]. However, reading and notating new music can be difficult for BLV users[21], and BLV music learners face complex obstacles due to their learning environment and music teachers[22].

3 Interview Insights and Design Concepts

Through interviews with three first-grade music teachers at BLV school, we identified the requirements and teaching methods in music education. The music curriculum encompasses instruction in music styles, braille notation, vocal singing skills, and instrumental playing techniques. These topics are systematically taught under the Music Curriculum Standards [4]. Teachers primarily employ listening and singing as teaching methods, including sing-along practice, audio instruction, and tactile feedback. Teachers noted that several first-grade students demonstrated musical talent after just half-semester. They can accurately address pitches, reproduce rhythms, and possess a sense of musical styles. Additionally, teachers widely acknowledge the profound impact of music education on BLV children. Learning music not only enhances auditory perception but also boosts self-confidence. It is estimated that 30% to 50% of BLV graduates possess exceptional musical talents and pursue careers in music, such as teaching, music therapy, and instrument tuning. However, in a lack of official curriculum standards for preschool-aged children, teachers suggest emphasizing musical interests over developing professional skills. Consequently, this study will examine the following issues based on interview findings and suggestions.

1. For students with good musical talent, can we cultivate their interest in music in the preschool stage to fully unleash their musical potential?
2. How will design promote BLV children's learning and understanding of music through tactile perception?
3. How can we enhance the efficiency of the interaction process to foster cognitive integration between auditory memory and tactile feedback in BLV children?

4 Prototype Design

The core concept of this prototype design is using modular blocks with different volumes to present music notes so that BLV children can build a logical bridge between music and tactile sensory experience. This prototype is designed with a 3D printed model and employs Arduino technology for interactive functions. Throughout the design iteration process, prototype design was guided by ergonomics and human-computer interaction principles and considering the mental model of BLV children. The volume, weight, and shape of the block have been adjusted several times to ensure accessibility and usability. We also deliberated on whether to engrave Braille onto the blocks to offer additional information, although this may lead to confusion and challenges for younger users. Consequently, any Braille or symbolic information has been excised from the prototype.

4.1 Model 1 and Model 2

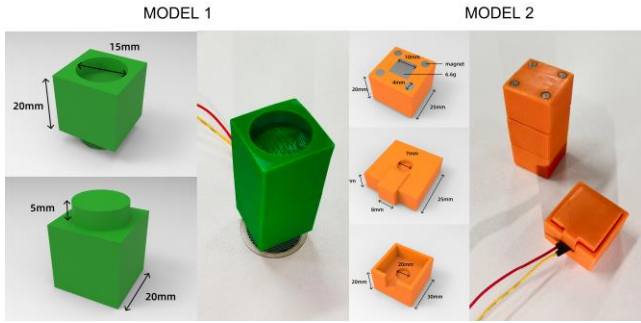


Figure 1: Model 1 and Model 2

The structure of Model 1, as depicted in Figure 1, comprises a vertical rectangular column and a base with a cylindrical nested structure. Rectangular prisms of varying heights represent distinct musical scales, with taller prisms indicating higher pitches. The Arduino components used in the prototype include Arduino Mega 2560 KEYES, Resistive Thin Film Pressure Sensor/Thin Film Pressure Sensor, and Mini MP3 Player Module. The Resistive Thin Film Pressure Sensor infers and retrieves music notes by measuring the weight of the block, but the precise qualitative assessment of the outcomes is less than ideal in practical use. Additionally, children have difficulty differentiating block differences through touch, which may impact their comprehension of syllable variations.

Therefore, Model 2 (shown in Figure 1) uses magnetic attraction to assemble the building blocks and adds an equal amount of counterweight to each, resulting in a final weight of about 15.7-15.9 grams. This stacking of equally sized building blocks aligns with children's mental model of ascending pitch, and the Resistive Flexible Thin Film Pressure Sensor C20B significantly enhances the accuracy of weight measurement. However, the added counterweights bring the total weight of the 7 blocks to approximately 110 grams, which may be too heavy for young children and could cause accidental injuries from falling blocks. Furthermore, when children notice a difference between the ideal pitch and the actual pitch, they often add blocks directly on top of those already placed. This additional process can cause the pressure sensor's fault recognitions, then affect the system's assessment.

4.2 Model 3

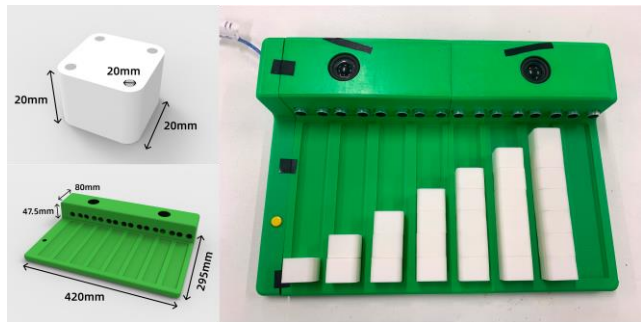


Figure 2: Model 3

Similar to Model 2, Model 3 also uses magnets to assemble blocks, enabling children to select the number of blocks according to the musical notes they wish to convey. Firstly, the weight of each block was reduced to 12.3g. Secondly, the blocks are now placed parallel to the desktop to prevent them from falling. The entire operational procedure has been streamlined. When the blocks are positioned at the bottom edge of the tray, the HC-SR04 Ultrasonic Distance Sensor can determine their height and play the corresponding notes directly. Additionally, the device adds a play button and the system can play eight notes in the order of the block arrangement to create a melody.

5 Initial Feasibility Test and Results



Figure 4: Children A and B in the Initial Feasibility Test

5.1 Participants and Test Contents

The research group invited two preschool children, Children A and B (as depicted in Figures 4), to participate in the initial feasibility test. Both children participated voluntarily and reviewed the test content with their parents before the test. The children's teacher monitored the test and provided necessary supervision. Before the test, the children were informed that they could immediately stop and withdraw if they felt any discomfort or disliked part of the test. To examine the usability and accessibility of NoteBlock for preschool children with minimal or no musical background, we selected two children who met our criteria. Child A, a 7-year-old male, has limited knowledge of musical notes. Child B, a 5-year-old male, has never studied music, though his brother plays the piano at home. Both have partial vision loss and low vision.

The test will consist of four steps. First, researchers will have children distinguish building blocks by touch. Second, the researchers will set up two block sets and have the children identify the higher and lower notes by hearing them. In the third step, the researchers will provide the children with two sets of blocks and play corresponding musical notes, asking if the pitch they hear matches the size of the blocks. For the final step, the researchers will compose a melody of eight scales using blocks and play it. Children can either press the play button to listen repeatedly or touch the building blocks to experience the variations.

5.2 Initial Feasibility Test Results

The results of the first three steps of the feasibility test are shown in Table 1 to 3. Firstly, children easily distinguished the sizes of

blocks through touch (with 100% accuracy of each) and also freely divided and reassembled the blocks easily. However, the two children demonstrated differing performances in addressing higher and lower notes. During the test, they heard 8 sets of distinct musical notes and 4 same notes played twice, with Child A correctly identifying 56.25% and Child B correctly identifying 50%. Both children could effectively address the differences in the referenced musical notes by touching and comparing the volume of the blocks, achieving a 75% accuracy rate for both. Finally, after the feasibility test, the children exhibited curiosity in interacting with the prototype and learning musical notes.

Table 1 Test 1 Content and Results

Distinguish the sizes of two different block sets									
	+/-	3	Results	+/-	2	Results	+/-	1	Results
	Blocks			Blocks			Block		
Child A	2 vs 5		T	2 vs 4		T	4 vs 3		T
	6 vs 3		T	4 vs 6		T	1 vs 2		T
	4 vs 1		T	5 vs 3		T	5 vs 6		T
Child B	5 vs 2		T	4 vs 6		T	6 vs 5		T
	4 vs 7		T	3 vs 5		T	3 vs 2		T
	6 vs 3		T	7 vs 5		T	4 vs 3		T

Table 2 Test 2 Content and Results

Address the higher and lower musical notes								
	+/-	2	Results	+/-	1	Results	Same Note	Results
	Scales			Scales				
Child A	A vs F		T	G vs A		T	G	T
	E vs G		F	E vs D		F	A	T
	C vs E		T	F vs G		T	F	T
Child B	G vs B		T	D vs E		F	E	T
	B vs G		F	A vs B		F	C	F
	A vs F		T	D vs C		F	F	T
	F vs D		T	G vs F		T	D	T
	E vs G		T	F vs G		T	A	T

Table 3 Test 3 Content and Results

Do the block sets match the notes				
	Music Notes	Block sets	Match (Yes / No)	Results
Child A	C vs E	1 Block vs 3 Blocks	N	F
	D vs C	2 Blocks vs 1 Blocks	Y	T
	E vs G	3 Blocks vs 5 Blocks	Y	T
Child B	D vs E	2 Blocks vs 3 Blocks	Y	T
	G vs A	5 Blocks vs 6 Blocks	Y	T
	E vs G	3 Blocks vs 5 Blocks	Y	T
	F vs G	4 Blocks vs 5 Blocks	N	F
	F vs A	4 Blocks vs 6 Blocks	Y	T

Besides, a few issues in the functional expression and interactive operation processes were revealed. Firstly, functional failures included errors in block recognition, external sounds, or sound malfunctions. Secondly, operational issues involved false

triggering due to sliding and placing building blocks, sensor malfunctions triggered by hand obstructions, and errors due to incorrect stacking of blocks. Additionally, children frequently pushed the blocks back and forth within the track during placement, which impacted the accurate determination of the block heights. Lastly, the close spacing between tracks meant that children could touch a block from a previous track while placing the next one, leading to displacement and affecting the overall playback effect.

6 Future Work

NoteBlock is designed to increase children's musical interest through tactile experiences, especially through activities focused on note recognition and simple composition. Although the children engaged effectively and demonstrated confidence in the tests, actual operations rarely resulted in issues like accidental touches, incorrect block placements, or incorrect operating procedures. Moreover, the prototype still exhibits some clear limitations. Following discussions with three music teachers, the following suggestions were put forward: 1. Enhance pitch contrast across multiple octaves. 2. Incorporate operational feedback or error alerts to prompt students to perform correctly, while distinguishing these reminders from actual note play. 3. Introduce a subsequent verification feature to enable students to practice correctly singing notes and tones. Additionally, these two children suggested incorporating tones from various instruments.

7 Conclusion

Through case studies and exploring mental models of music learning in preschool children, we found that tactile stimulation helps BLV children in better understanding of abstract music concepts, such as notes and scale variations. NoteBlock is proposed as a block system as an interactive prototype for BLV or other young children to enrich their musical experiences. The research group organized interviews with three teachers at BLV School and engaged two preschool children in the initial feasibility test. The test revealed that tactile stimulation and auditory interaction can significantly boost BLV children's interest in music learning, yet the prototype design also presents inherent limitations. Future research will concentrate on refining the prototype and exploring additional features.

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